



STN-AVI-CON-02

Connectors Standard

Avionics Department



Table 1. Document History

Version	Comments	Date	Author	Approved by
V0.01.0	Created document	28/06/26	RH	

Table 2. Approval

Role	Name	Signature	Date
Avionics Lead	Raphael Ho	RH	15/07/2026
Avionics Lead	Juin Jieh Chew	JJC	15/07/2026
ADCS Lead	Kalani Gough	KG	25/07/2026
Systems Engineering Lead			
Supervisor / Advisor			

Table 3. Contact Details

Name	Discord	Email
Raphael Ho	Raph.h	22197696@student.curtin.edu.au



Table of Contents

Document Purpose	5
Out of Scope	5
General Information	5
EPS Power Connectors	6
High Power Connector	6
Orientation v1 (NRFND)	7
Orientation v1 Incomer Connection (Same Side Cable)	7
Orientation v1 Outgoer Connection Orientation (Same Side Cable)	8
Orientation v2	8
All Connection Orientations	8
Medium Power Connector	9
Deviations	10
Signal & Low Power Connector	11
16 Pin FPC Connector	11
MCU / Controller Orientation	11
Peripheral / Sensor Orientation	12
Protocol Orientations	13
0.1" Connector	14
Deviations	14



List of Tables

Table 1. Document History	2
Table 2. Approval	2
Table 3. Contact Details	2
Table 4. Related Documents	5
Table 5. Nomenclature / Acronyms	5
Table 6. High Power Connector Specifications	6
Table 7. Multiple Connection Combinations	7
Table 8. Medium Power Connector Specifications	9
Table 9. MCU / Controller Orientation	11
Table 10. Peripheral / Sensor Orientation	12

List of Figures

Figure 1. Example Image of a 504050 Series Connector	6
Figure 2. Orientation v1 of the High Power Connector (Incomer)	8
Figure 3. Orientation v1 of the High Power Connector (Outgoer)	8
Figure 4. Orientation v2 of the High Power Connector	9
Figure 5. Orientation of the Medium Power Connector	10
Figure 6. Example Image of a 505110-1692 FPC Connector	11
Figure 7. Connection between MCU & Peripheral / Controller & Sensor	12
Figure 8. UART Connection Diagram	14



Table 4. Related Documents

Document Number / Link	Name
Avionics Inventory.xlsm	Inventory
STN-AVI-REC-01	Recommended Avionics Components

Table 5. Nomenclature / Acronyms

Acronym	Meaning
CubeSat	Cube Satellite
NRFND	Not Recommended For New Designs
PAST	Perth Aerospace Student Team
PCB	Printed Circuit Board

Document Purpose

The purpose of this document is to outline standard connectors PAST uses, and how the connectors should be wired up to prevent interface incompatibility or hazards.

This document is for anyone designing a PCB that may require interfacing between other boards / PCBs.

Out of Scope

Connectors that do not interface to other systems of the CubeSat, such as:

- Connecting to a computer
- Connecting to power – Mains / Power Supply / Battery Cell
- Connectors for debug purposes

General Information

EPS power connectors refer to connectors that are designed to supply dedicated power to a PCB.

FPC connectors are designed for signal and power (up to 2A). Some systems have their power pins disconnected to prevent drawing power through them.

Gold plated cables / connectors are to connect only to gold plated cables / connectors. Tin plated cables / connectors must not connect to gold plated cables / connectors as this can cause corrosion and premature damage.



It is not recommended to use tin cables / connectors in spacecraft and is a risk that the Systems Lead must acknowledge and approve. The use of tin-plated connectors may additionally require launch provider approval.

EPS Power Connectors

The EPS power connectors are determined based on the power requirements. The connectors should be specified such that the rating is higher than the maximum possible current draw, such as from the batteries, or from the buck converters.

In a 2P2S system, the maximum current from the batteries can peak to 10A, however the current battery protection circuit (BOOM V1) only allows up to 4A per battery, limiting the current draw to 8A.

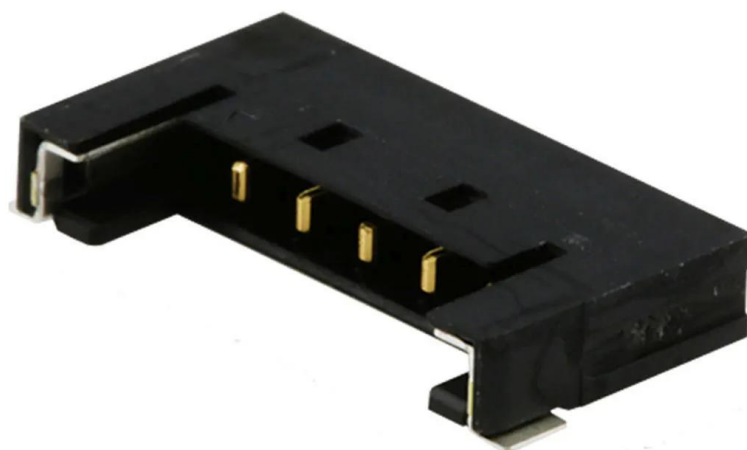


Figure 1. Example Image of a 504050 Series Connector

High Power Connector

The high-power Connector is the Molex Pico-Lock 8 Pin connector which is rated up to 8A of current and is provided in two different connection orientations. These are used exclusively between the batteries and power distribution boards and for testing the batteries.

In theory, once the load of a satellite has been determined, a Medium Power Connector could be used if suitable to save cost & space.

Table 6. High Power Connector Specifications

Specification	Value	Notes
Current	8A	Maximum continuous



Connector Part Number	504050-0891	
Cable Size	24 AWG	
Cable Part Number	15132-0801	Multiple other part numbers & lengths available 
Use	EPS internal connections	

Orientation v1 (NRFND)

This is the original orientation for the connectors which have positive and negative ends on separate sides. This allows compatibility with BINAR, however requires connectors to mirror the connections to prevent the positive side from connecting to a negative side when using a same side cable. When using a opposite side cable, the connector must not mirror the other connector.

Care must be taken to not connect the wrong orientation otherwise a reverse voltage will occur which may cause damage to hardware.

Table 7. Multiple Connection Combinations

Orientations	Connectors Not Mirrored	Connectors Mirrored
Same Side Cable	NOT ALLOWED	ALLOWED
Opposite Side Cable	ALLOWED	NOT ALLOWED

This design is NRFND but is shown for backwards compatibility and awareness.

Orientation v1 Incomer Connection (Same Side Cable)

This connector is used for the incomer (boards receiving power)

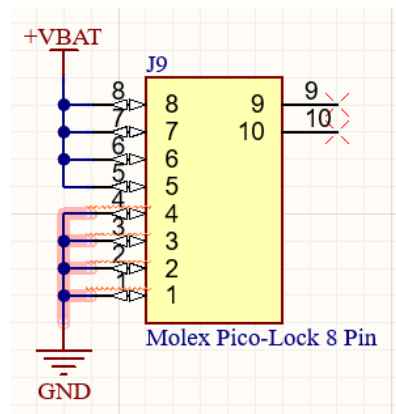


Figure 2. Orientation v1 of the High Power Connector (Incomer)

Orientation v1 Outgoer Connection Orientation (Same Side Cable)

This connector is used for the outgoer (boards providing power)

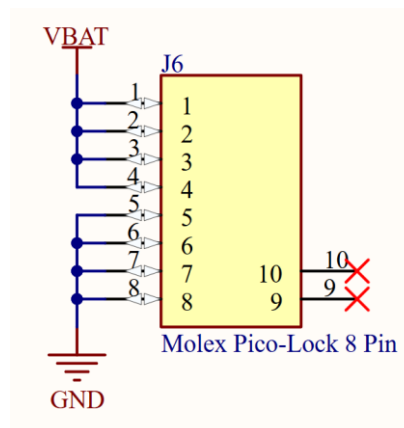


Figure 3. Orientation v1 of the High Power Connector (Outgoer)

Orientation v2

This is the current orientation of the High Power Connector, which the positive end is in the center of the connector, and the negative ends are on the outside edges. This allows a same side or opposite side connector to connect to a mirrored or non-mirrored connector, removing any incompatibility with different boards and cables.

All Connection Orientations

There are no requirements for the specific pin orientation of the connector, other than the center pins are positive and others are negative due to the symmetry of the connector.

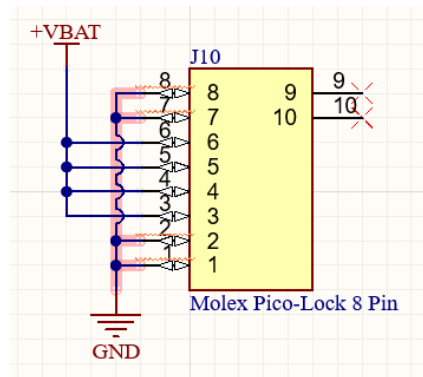


Figure 4.Orientation v2 of the High Power Connector

Medium Power Connector

There is currently only one orientation of the Medium Power Connector due to the symmetry of the connector. The connection is similar to Orientation v2 of the High Power Connector, however with half the pins used instead.

A 4 pin connector allows power to still be delivered (at a reduced capacity) if one cable is to be damaged or have a faulty connection.

Table 8. Medium Power Connector Specifications

Specification	Value	Notes
Current	4-6A	Maximum continuous Connectors are rated to 6A; however cables may not be rated to 6A
Connector Part Number	504050-0491	
Cable Size	24 AWG	
Cable Part Number	15132-0406	Multiple other part numbers & lengths available 
Use	EPS to major subsystem connections	Major subsystems include any system that requires to be electrically isolated to reduce noise or power loss, or may carry a current higher than 2A

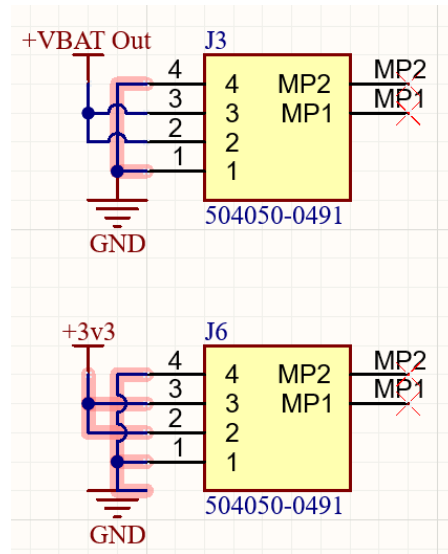


Figure 5. Orientation of the Medium Power Connector

Deviations

Deviations for the EPS Power Connectors are not recommended and will require approval from the Avionics Lead & Systems Lead. When approving a deviation, considerations with compatibility, usability, and cost to benefit have to be taken into account.



Signal & Low Power Connector

16 Pin FPC Connector

The main signal & power connector used is the 16 Pin FPC Connector. Due to the asymmetry of the connector, and the use of same side cables, each pair of connectors must be flipped each.

The orientation is specified to be backwards-compatible with previous designs & compatible with BINAR.

Each pin has a current rating of 0.5A, so each connector with the specified orientation has a current rating of 2A.

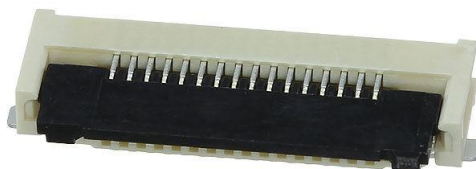
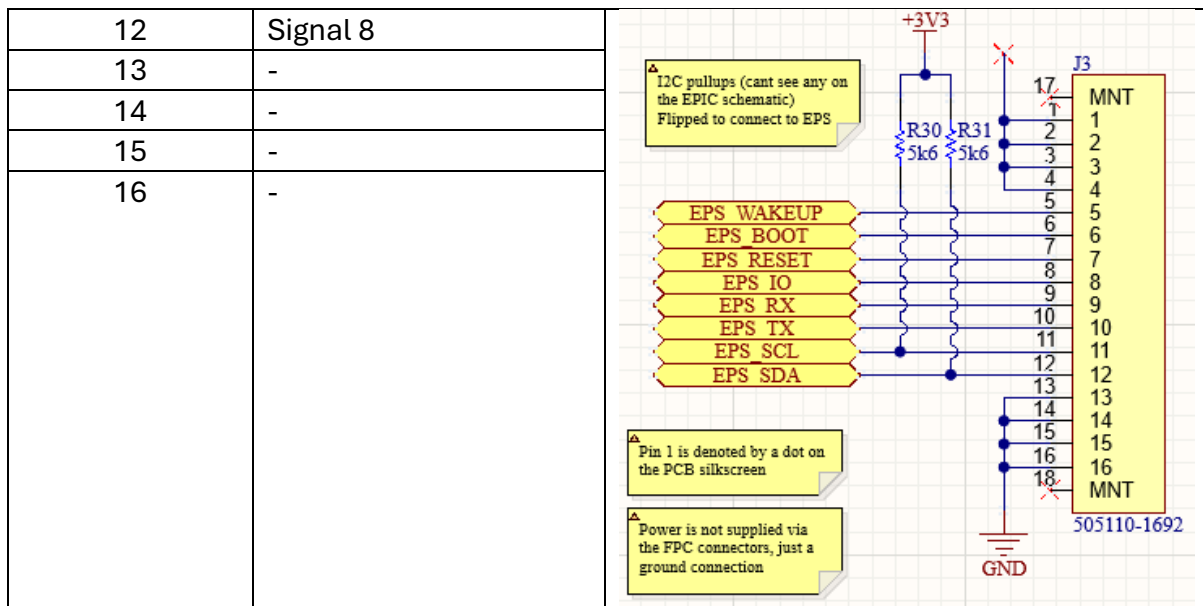


Figure 6. Example Image of a 505110-1692 FPC Connector

MCU / Controller Orientation

Table 9. MCU / Controller Orientation

Pin	Use	Schematic
1	+	
2	+	
3	+	
4	+	
5	Signal 1	
6	Signal 2	
7	Signal 3	
8	Signal 4	
9	Signal 5	
10	Signal 6	
11	Signal 7	



Peripheral / Sensor Orientation

Table 10. Peripheral / Sensor Orientation

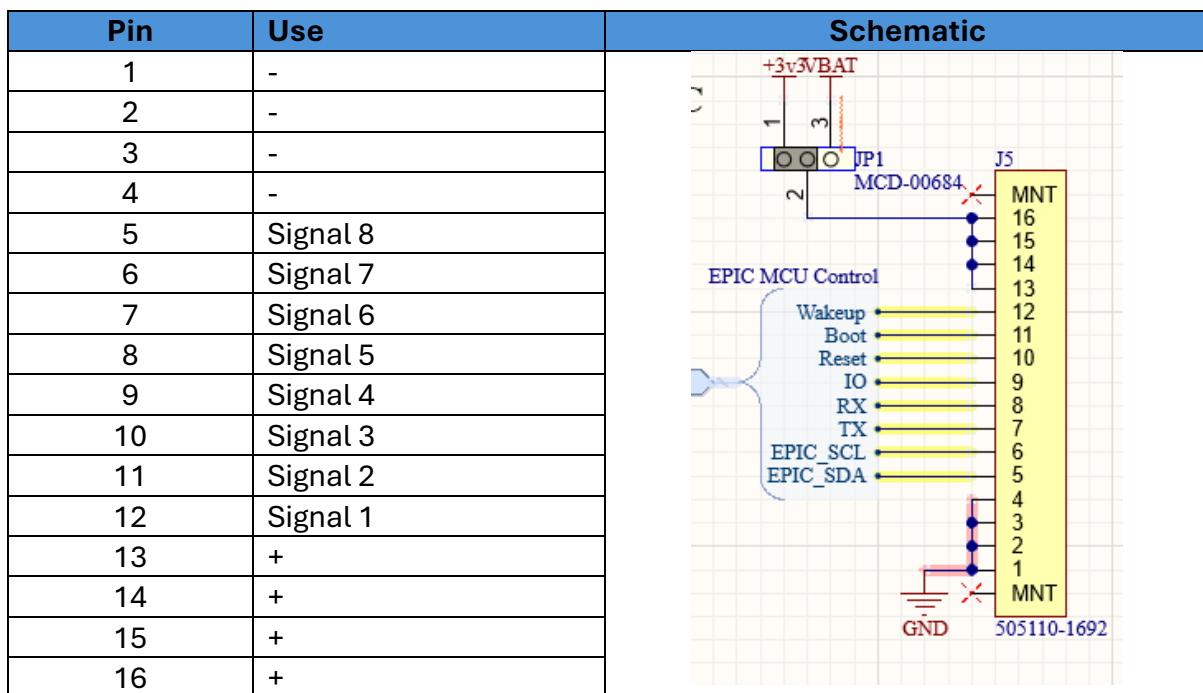
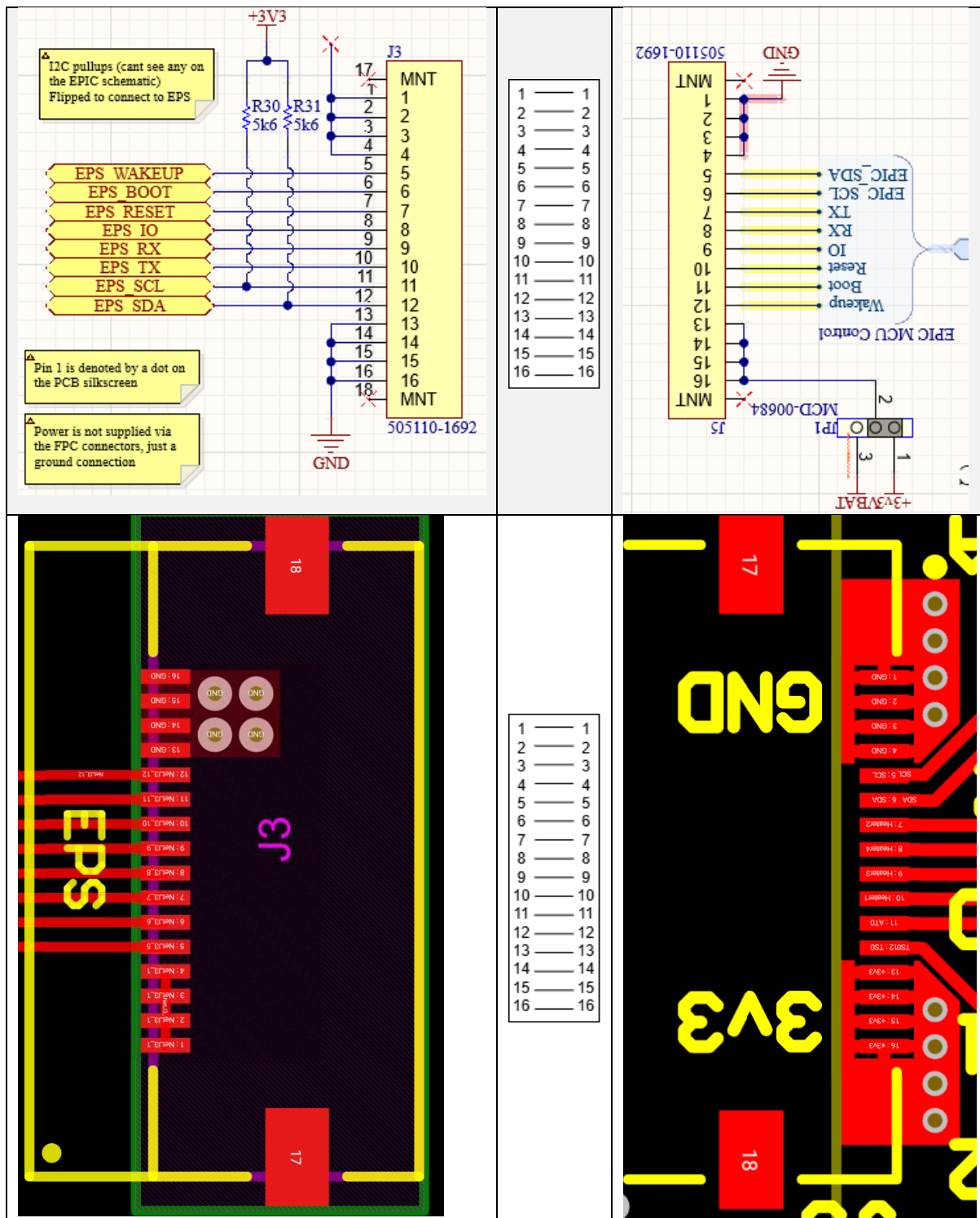


Figure 7. Connection between MCU & Peripheral / Controller & Sensor

MCU / Controller	Cable	Peripheral / Sensor
------------------	-------	---------------------



Protocol Orientations

Protocols such as UART require the TX and RX pins between devices to be swapped, requiring an orientation change. The label on each device should specify the device's connection rather than the connection of the connecting device.



Some STM32 devices allow the TX and RX pins to be swapped in software, allowing an incorrect orientation to still be functional.

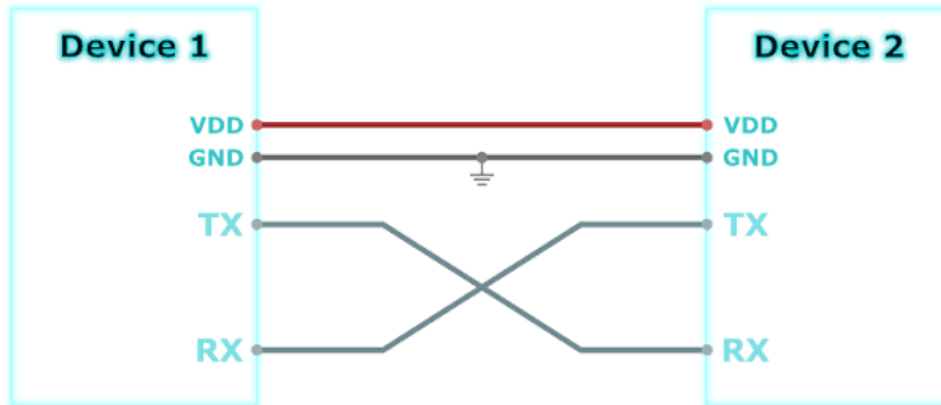


Figure 8. UART Connection Diagram

0.1" Connector

0.1" Connector can be used, it is recommended to have GND & +VCC pins on each board for debug purposes, and for signals.

Devices drawing a current above 1A should not use this connector.

Deviations

Deviations for signal and low power are allowed if there is a valid reason to not use other specified connectors. Reasoning **MUST** be documented along with sufficient research.